# Tuning the group identification

How the delay $\delta$, the false discovery rate $q$ and the duration $\tau^*$ were chosen on the FFVL archive

Matteo Di Sante

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Companion of the note *Group flights in free flight* (same folder)

**The choice.** Two pilots are in contact when one of them passes within $\varepsilon_h = 500$ m and $\varepsilon_z = 300$ m of where the other was at most $\delta = 2$ min before or after; headings are compared only when both glide ($\theta_{\max} = 45°$) and gaps shorter than $g = 120$ s are joined. A contact counts when it lasts at least $\tau^* = 38.1$ min, the shortest duration at which chance explains at most $q = 0.1$ of the contacts kept. With these values 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes.
**Why.** The delay was measured: real pairs have 3.7 to 5.7 times the correspondences of chance within 2 min, less beyond (Section 3). $\tau^*$ is not set by hand: it follows from $q$ and from the null model (Section 2); $q$ is the trade-off between how many groups we see and how many of them are chance (Section 4).
**Where.** `/Volumes/SSD_DISANTE/xc-lab/group-identification/runs/d120_eh500_ez300_th45_g120_q0.1`, with a `PARAMETERS.md` that lists every parameter.

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## 1 What is tuned

The method of the companion note has three kinds of parameters (Table 1). The *tolerances* say when two positions match: $\delta$, $\varepsilon_h$, $\varepsilon_z$, $\theta_{\max}$, $g$. The *duration* $\tau^*$ says how long a match must last

to be a contact; it is not chosen directly but computed from the share $q$ of chance contacts we accept. The other *settings* make the computation exact or fast and are fixed by physical bounds.

| Parameter | Value | How it was set |
|---|---|---|
| $\delta$, largest delay | 2 min | measured: delay profile, Section 3 |
| $q$, accepted false discovery rate | 0.1 | trade-off, Section 4 |
| $\tau^*$, shortest contact | 38.1 min | computed from $q$ and the null model, Section 2 |
| $\varepsilon_h$, $\varepsilon_z$ | 500 m, 300 m | definition of "the same air" (companion note); sensitivity in Section 5 |
| $\theta_{\max}$ | 45° | no effect on the result (Section 5) |
| $g$, longest gap joined | 120 s | starting value of the note; sensitivity in Section 5 |
| $\Delta$, $h_{\max}$, $R_{\text{dep}}$, $v_{\max}$, $R_s$, $r_{\min}$ | see Section 6 | physical bounds and exactness |

**Table 1:** The parameters and how each was set.

## 2 Chance contacts and $\tau^*$

Two pilots who never interact still meet: they take off from the same site, follow the same ridges and valleys and climb in the same house thermals. The null model of the companion note measures how often. For pairs of flights of the same day whose takeoffs are less than $R_s = 10$ km apart, one flight is replaced by its copy shifted in time by more than $2\delta$, so that every real correspondence is destroyed while routes and thermals stay the same. The contacts of these surrogate pairs are chance. On date $d$, with $\rho_{0,d}(\tau)$ the surrogate intervals lasting at least $\tau$ per second of surrogate exposure and $H_d$ the exposure of the real pairs (the time both flights fly after departure), chance gives $\sum_d \rho_{0,d}(\tau) H_d$ intervals lasting at least $\tau$. With $N(\tau)$ the real ones,

$$\widehat{\text{FDR}}(\tau) = \frac{\sum_d \rho_{0,d}(\tau)\, H_d}{N(\tau)}, \qquad \tau^* = \min\{\tau : \widehat{\text{FDR}}(\tau) \le q\}. \tag{1}$$

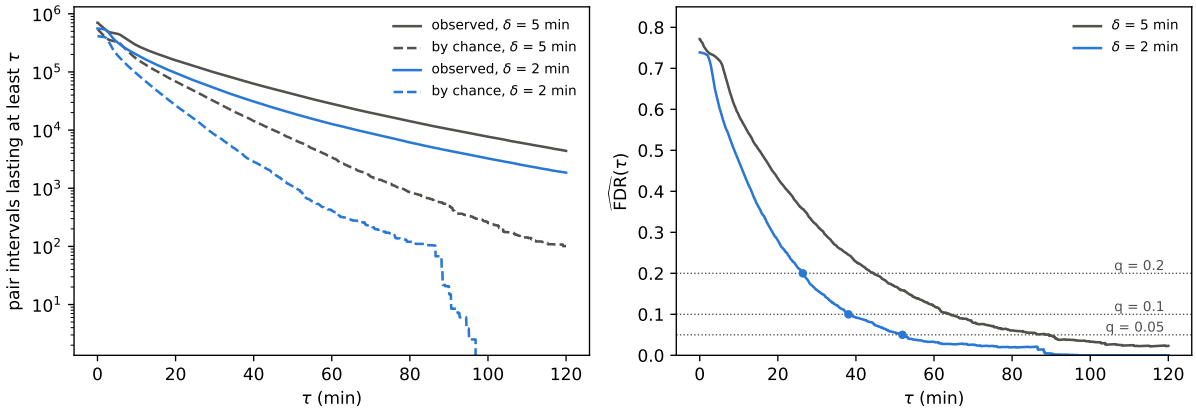



**Figure 1:** Left: real pair intervals lasting at least $\tau$ (solid) and those chance alone gives (dashed), with $\delta = 5$ min (grey) and $\delta = 2$ min (blue), on the whole archive. Right: their ratio, the false discovery rate of Eq. (1); dots mark $\tau^*$ for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

Figure 1 is the central fact of the tuning. Most contacts are chance: 74% of all pair intervals with $\delta = 2$ min (77% with 5 min), 47% of those lasting 10 min, 16% of those lasting 30 min. Short contacts therefore say little; only long ones are mostly real. This is why $\tau^*$ is tens of minutes: it is the duration at which the real contacts outnumber chance by the factor $1/q$.

Resampling the dates 200 times moves $\tau^*$ by a few minutes (Table 2), so the curve, not the noise of the surrogates, sets it.

# 3 The delay $\delta$

## 3.1 The delay profile

A wider $\delta$ catches pilots who follow farther behind, but it also adds chance correspondences. To see where the signal ends, we counted, for every delay $l$, the correspondences $(t, t+l)$ of the real pairs and of the surrogate pairs, per second of exposure, with a window of 15 min on 300 random dates (58 910 h of real exposure):

$$R(l) = \frac{\text{real correspondences at delay } l \text{ per second of exposure}}{\text{chance correspondences at delay } l \text{ per second of exposure}}. \tag{2}$$

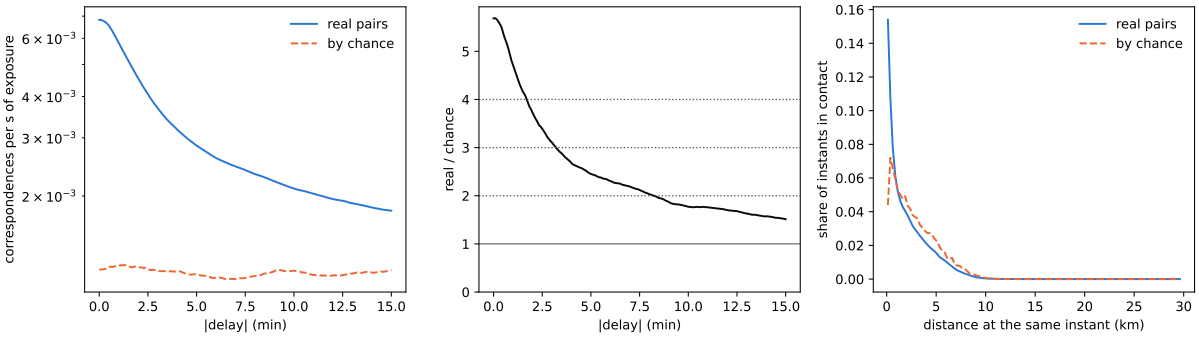



**Figure 2:** The delay profile on 300 dates with a 15 min window. Left: correspondences per second of exposure at each delay, real and by chance. Middle: their ratio $R(l)$ of Eq. (2), with the levels 4, 3 and 2. Right: the distance between the two pilots at the same instant when one is on the path of the other.

Chance is flat in the delay, as it should: a shifted copy has no preferred delay. The real pairs peak at a few seconds, where they have 5.6 times the correspondences of chance, and decay: $R$ falls below 4 at 105 s, below 3 at 195 s and below 2 at 500 s, and is still 1.5 at 15 min (Figure 2). The excess has two parts. The fast one, within 2–3 min, is pilots flying together. The slow one, which has not ended at 15 min, is pilots who use the same thermals at about the same time of day without following each other: the shift of the null model, more than 30 min with this window, removes that synchronisation too. A rule such as "the delay that holds 95% of the excess" would pick 13 min, because it counts the slow tail; that is why we read the ratio instead.

## 3.2 The delay inside the long contacts

The contacts that survive $\tau^*$ tell the same. Figure 3 shows, for the pair intervals of the run with $\delta = 5$ min, the median delay between the two pilots inside each interval. Short intervals pile up at the edge of the window, 300 s: 29% of those under 10 min, the signature of chance matches, which are spread over all delays and are found at the largest ones. Long intervals do not: those of at least 90 min have a median delay of 25 s, 90% of them within 2 min, and only 6 of 10 734 at the edge.

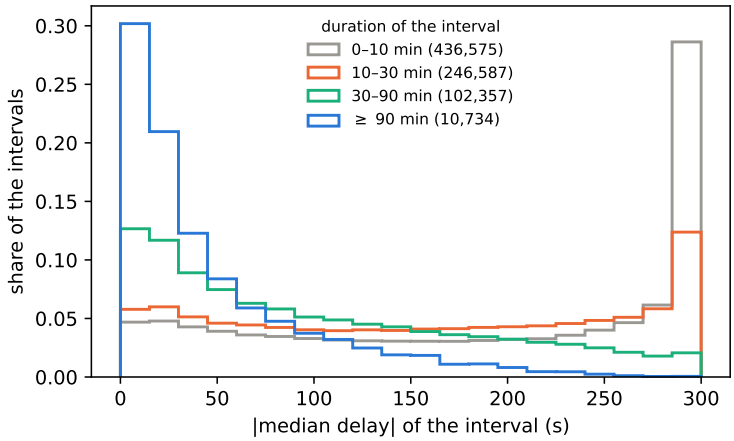



**Figure 3:** Median delay between the two pilots inside each pair interval, by the duration of the interval ($\delta = 5$ min, whole archive).

### 3.3 The choice

We take $\delta = 2$ min: within it the real correspondences are at least 3.7 times chance, and it contains nine in ten of the long contacts. Shortening the delay also shortens $\tau^*$, since fewer chance matches enter: from 89.5 to 51.9 min at $q = 0.05$ (Table 2).

The delay also bounds how far apart two pilots in contact can be at the same instant: at most $\varepsilon_h + v_{\max}\delta \approx 0.5 + 27.8 \times 120/1000 \approx 3.8$ km. A separation of 6 km, the largest we want to call a group, is never reached by one pair; a group, a chain of pairs, can be wider. Even with the 15 min window, 58% of the instants in contact have the two pilots within 2 km and 94% within 6 km (Figure 2, right).

## 4 The false discovery rate $q$

$q$ is the share of chance contacts we accept among those we keep. A smaller $q$ keeps only very long contacts and few groups, all of them real; a larger $q$ sees more groups, with more chance among them. Table 2 and Figure 4 give the trade-off.

| $\delta$ | $q$ | $\tau^*$ (min) | 5–95% over dates | para in a group | hang in a group | episodes |
|---|---|---|---|---|---|---|
| 5 min | 0.05 | 89.5 | 79.5–91.5 | 8.9% | 2.1% | 6 090 |
| 2 min | 0.05 | 51.9 | 49.1–53.3 | 13.7% | 3.8% | 10 553 |
| 2 min | 0.1 | 38.1 | 37.0–41.1 | 21.7% | 6.9% | 17 169 |
| 2 min | 0.2 | 26.4 | 25.2–27.2 | 31.7% | 13.9% | 26 504 |

**Table 2:** $\tau^*$ and the groups it gives. Flights with a logger of at most 10 s and a position after departure: 154 598 paragliders and 6 039 hang gliders.

We take $q = 0.1$: at most one contact in ten is chance, and $\tau^* = 38.1$ min. Going to 0.2 would lower $\tau^*$ to 26 min but would let chance make one group in five.

## 5 Sensitivity to the other tolerances

The companion note asks for a plateau: values of the tolerances around which the result does not change. We changed one parameter at a time on 300 random dates, recalibrating $\tau^*$ each

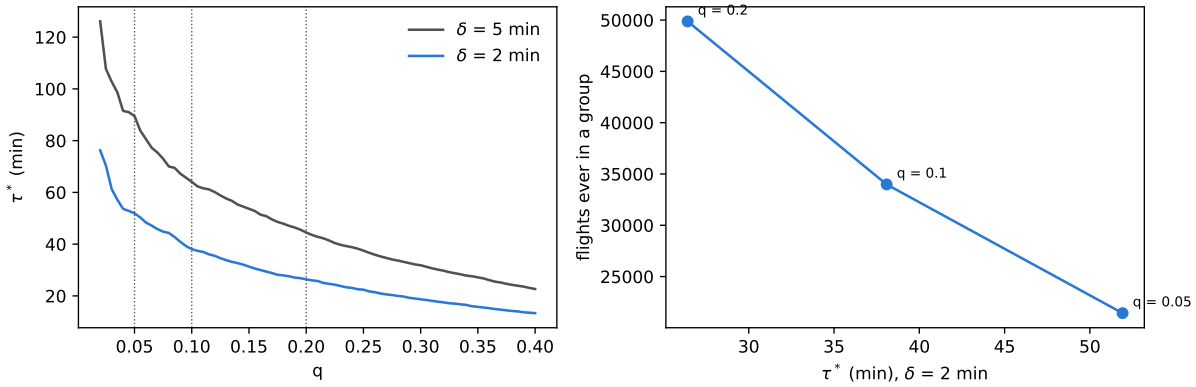



**Figure 4:** Left: $\tau^*$ as a function of $q$, for the two delays. Right: flights ever in a group against $\tau^*$, for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

time, with $\delta = 5$ min (Table 3). $\theta_{\max}$ changes nothing: contacts in glide are few. The others move the fraction of flight time in a group, $f_G$, by up to a factor three, mostly through $\tau^*$: there is no plateau, so the tolerances are definitions to declare rather than values the data single out. On 300 dates $\tau^*$ is noisy (126 min with the starting values against 89.5 on the whole archive, and six settings stop at the same surrogate interval); a conclusive check needs the whole archive for each setting, about one hour each. It was not repeated with $\delta = 2$ min.

| Setting ($\delta = 5$ min unless stated) | $\tau^*$ (min) | $f_G$ | episodes | flights in a group |
|---|---|---|---|---|
| starting values | 126.1 | 0.034 | 98 | 210 |
| $\varepsilon_h = 250$ m | 75.1 | 0.058 | 253 | 537 |
| $\varepsilon_h = 1000$ m | 126.8 | 0.065 | 162 | 372 |
| $\varepsilon_z = 150$ m | 126.1 | 0.020 | 61 | 125 |
| $\varepsilon_z = 600$ m | 126.1 | 0.044 | 122 | 268 |
| $\delta = 2$ min | 42.6 | 0.106 | 600 | 1 170 |
| $\delta = 10$ min | 128.4 | 0.062 | 153 | 365 |
| $\theta_{\max} = 30°$ | 126.1 | 0.034 | 97 | 208 |
| $\theta_{\max} = 90°$ | 126.1 | 0.034 | 98 | 210 |
| $g = 60$ s | 75.8 | 0.087 | 332 | 732 |
| $g = 300$ s | 126.1 | 0.055 | 141 | 318 |

**Table 3:** One parameter at a time on 300 random dates (7 051 flights), $q = 0.05$.

## 6 Fixed settings

- $\Delta = 5$ s, the common grid: a pilot is at most 69 m from a grid position at 100 km/h, against $\varepsilon_h = 500$ m.
- $h_{\max} = 10$ s: positions are interpolated only between fixes at most 10 s apart; the flights of coarser loggers are left out of the statistics.
- $R_{\text{dep}} = 3$ km: contacts around the takeoff, before the first exit from this disc, are not counted.
- $v_{\max} = 100$ km/h, the speed of the candidate filter: pieces of track end at every faster step, so the filter never misses a correspondence (checked against every pair of instants on synthetic days).
- $R_s = 10$ km, $r_{\min} = 2\delta + \Delta$, at most 200 surrogate pairs per date: the null model of Section 2.

## 7 Limits

- The null model shifts tracks by more than $2\delta$, which also removes the synchronisation of pilots with the weather of the day. What it calls real therefore includes pilots who share the same thermal at the same time without following each other: the "same air" notion of the companion note, not proof of interaction.
- $\varepsilon_h$ and $\varepsilon_z$ are definitions; a distance profile, the analogue of Figure 2 for the horizontal and vertical gap, would measure them as the delay was measured.
- The plateau of Section 5 is on 300 dates and with the old delay.
- The data are the cleaned tracks of the old repository (soaring, tag `legacy-final`); the phases needed by the heading condition exist on 1 Hz flights only.